

Implementation Plan

Version: 0.1 (draft). Timeline assumes 1–2 developers working full time on the proposed stack (Next.js + Supabase). Durations are estimates; adjust once team size is known.

1. Delivery Strategy

- Build the thinnest end-to-end flow first (post → claim → collect → report), then layer the map, score, and public features on top.
- Ship to a small pilot early (about week 8) and tune the scoring with real data.
- Keep every feature behind the MVP scope in the PRD; defer everything else.

2. Phases

Phase 0: Setup (Week 1)

- Create repo, Next.js + TypeScript + Tailwind project, linting and formatting.
- Create Supabase projects (staging, production); enable PostGIS.
- Set up CI, environment variables, Sentry.
- Define design tokens (colours, type, spacing) from the UI/UX doc.
- **Done when:** a "hello world" PWA deploys to staging and installs on a phone.

Phase 1: Accounts, Roles, Admin Approval (Week 2)

- Auth (email/password or phone OTP), `profiles`, roles.
- Hotel and NGO registration forms with map pin picker and document upload.
- Admin approval queue; approval emails.
- RLS policies for all tables created so far.
- **Done when:** a hotel and an NGO can register and be approved by an admin.

Phase 2: Listings and Claiming (Week 3)

- Create listing form with defaults, templates, "repost last", safety checklist.
- NGO list view of nearby open listings (PostGIS radius query), filters.
- `claim_listing` function with atomic claim; claim and release flows.
- Auto-expiry job; volunteer invite and assignment.
- In-app notifications; web push setup.
- **Done when:** a hotel posts, an NGO is notified, claims, and assigns a volunteer.

Phase 3: Map and Live Tracking (Weeks 4–5)

- Leaflet map with urgency-coloured pins, clustering, bottom sheet.
- Pickup states and trip screen; `watchPosition` with throttling; wake lock.

- `location_pings` writes and Realtime subscription on the hotel/NGO side.
- Routing API for route line and ETA; stale-location handling.
- Simulated-movement test script.
- **Done when:** a hotel can watch the volunteer's marker move on a real phone during a test trip.

Phase 4: Collection, Report and Flags (Week 6)

- Geofence check on "Collected"; photo capture and upload (private bucket).
- Completion report: portions, tags, left-behind; reminder jobs.
- Auto-flags and admin flag queue with Keep / Correct / Void and audit log.
- **Done when:** a full pickup can be completed and an admin can void a report.

Phase 5: Score, Leaderboard and Impact Passport (Week 7)

- Score calculation (SQL function or Edge Function), weekly snapshot job.
- Leaderboard views, tiers, categories, Most Improved, Streak.
- Hotel private dashboard; monthly PDF certificate; embeddable badge; QR card.
- Admin settings screen for weights and constants.
- **Done when:** seeded data produces a believable leaderboard, and certificates and badges render.

Phase 6: News and Public Experience (Week 8)

- Public sign-up, news feed, internal stories, city counter.
- RSS ingestion to drafts and admin approval.
- Follow and kudos; hotel public pages; share previews.
- **Done when:** a public user can sign up, read news, follow a hotel and send kudos.

Phase 7: Hardening and Pilot Launch (Weeks 9–10)

- Device testing on iOS Safari and Android Chrome; fix permission, camera, push, wake-lock issues.
- Security review: RLS audit, rate limits, signed URLs, input validation.
- Accessibility pass and performance check.
- Legal: terms of use, privacy policy, food-safety wording reviewed.
- Support email and complaint procedure ready.
- Onboard 3–5 hotels and 2–3 NGOs; training session for volunteers.
- **Done when:** the pilot is live with real pickups.

3. Pilot (Weeks 10–14)

- Run for about 4 weeks; review metrics weekly (claim rate, time to claim, report completion, repeat posting).
- Collect feedback from hotels, NGO volunteers and a few public users.
- Tune score weights, benchmark constants, notification radius and reminder timing.

- Decide on next steps: wider launch, WhatsApp/SMS alerts, independent volunteers, native app.

4. Milestones and Checkpoints

Milestone	Target	Demo-able output
M1: Accounts and approval	End of week 2	Register and approve hotel + NGO
M2: Core flow	End of week 3	Post, notify, claim
M3: Live map	End of week 5	Moving marker on a phone
M4: Report and moderation	End of week 6	Complete pickup, admin void
M5: Score and incentives	End of week 7	Leaderboard, certificate, badge
M6: Public features	End of week 8	News, follow, kudos
M7: Pilot launch	Week 10	Real pickups with partners

5. Task Breakdown by Area

Area	Key tasks
Front end	Design system, layouts, forms, map, sheets, dashboards, public pages, PWA
Back end	Schema and migrations, RLS, functions, jobs, notifications, PDF generation
Integrations	Routing API, web push, email, RSS, OSM tiles
QA	Test plan, device matrix, seed data, simulated trips
Ops	CI/CD, backups, monitoring, secrets, environments
Content	Tag list, safety checklist wording, terms, onboarding copy, first news stories
Partners	Recruit pilot hotels and NGOs, onboarding sessions, feedback loop

6. Team and Roles (suggested)

- 1 full-stack developer (core), 1 front-end/UI developer (optional but helpful).
- 1 product owner (you) for decisions, partner relationships, content.
- Part-time designer for wireframes and brand (or use the token set as is).

- Part-time legal review for terms and food-safety wording.

7. Risks to Watch During the Build

Risk	Early signal	Response
Live tracking unreliable on iOS	Test trips drop updates when screen locks	Emphasise wake lock; status fallback; "last seen" label
Hotels don't post after week 1	Falling weekly post count	Personal onboarding, reminders to kitchen staff, certificates and badges early
NGO reports late or skipped	Report-due flags rising	Auto-reminders, show the report step right after collection, trust score
Scope creep	New requests mid-phase	Park in a backlog; only revisit after pilot
Score feels unfair	Complaint emails	Review constants, void/correct, adjust smoothing

8. Post-MVP Backlog (not in scope now)

- WhatsApp/SMS alerts for NGOs.
- Independent volunteers and delivery routing across multiple pickups.
- Multi-language UI.
- Native apps.
- Connected kitchen scales or surplus forecasting.
- Corporate sponsorship and diner-points redemption at partner hotels.
- Multi-branch hotel chains.
- Dark mode.

9. Immediate Next Steps (this week)

1. Answer the open questions (listed with this document set) so the PRD and schema can be finalised.
2. Decide the stack (confirm Next.js + Supabase) and who will build it.
3. Choose a product name and register the domain.
4. Shortlist 3–5 pilot hotels and 2–3 NGOs and start conversations.
5. Draft the terms of use and food-safety checklist for legal review.
6. Begin Phase 0.